MIL-R-81407A (AS)
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Superseding
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15 March 1966

MILITARY SPECIFICATION

Recorder-Reproducer Set, Signal Data

AN/ALH-6(*)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall receive, monitor, and store analog and digital signals on magnetic tape.

* 1.2 Classification - The Recorder-Reproducer Set, Signal Data shall consist of the following items.

ITEM	TYPE DESIGNATION	APPLICABLE PARAGRAPH
Control, Recorder	C-7237/ALH-6	3.5.1
Recorder, Signal Data	RO-330/ALH-6	3.5.2
Converter, Signal Data	CV-2278/ALH-6	3.5.3
Converter, Signal Data	CV-2346/ALH-6	3.5.4
Converter, Signal Data	CV-2279/ALH-6	3.5.5
Recorder, Signal Data	RO-329/ALH-6	3.5.6

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

FSC 5835

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2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-23603	System Readiness/Maintainability; Avionics System Design, General Specification for

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Naval Air System Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of

STANDARDS

Military

MIL-STD-595	Colors
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS17322	Meter, Time Totalizing, 115 Volt, 400 Cycle

2.1.1 Availability of Documents

When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

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3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

*3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern.

(1) Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies shall be reliable. (see 6.5)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

*3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors, and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in Specification AR-5.

*3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5.

3.2.3 Modules - The electronic portions of the equipment shall be divided into maintenance modules (see Specification MIL-S-23603). Maintenance modules shall normally be considered repairable. The visual go-no-go warning requirements of MIL-S-23603 shall not apply.

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*3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction, and workmanship, unless otherwise specified herein. All equipments whose construction is not covered by MIL-C-6781 shall be painted lusterless black using paint number 37038 per Federal Standard 595; spare pins shall be provided on each connector, but not necessarily to the limits called for in MIL-E-5400.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 90 pounds.

3.3.2 Reliability

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 100 hours without the necessity for readjustments of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 2500 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 250 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The following units shall contain time totalizing meters in accordance with Specification MIL-M-7793.

Unit	Type of Meter
Recorder, Signal Data RO-330/ALH-6	MS-17322
Recorder, Signal Data RO-329/ALH-6	MS-17322

3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

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3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

* 3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781 (Type 1). The configuration of all control panels must be approved by the procuring activity prior to the preproduction testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-I-6181.

3.3.7 Provisions of Maintainability - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

* 3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC and 27.5 ± 0.5 VDC

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3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 1 equipment, except as modified herein:

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.3.11 Warm-up-Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed five minutes under standard conditions and fifteen minutes at extreme service conditions.

3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amount.

(1) AC Power (Three Phase), 115/200 V, 400 cps, Category "B" 400 VA

(2) DC Power 28 V DC, Category "B" at 0.8 amp

(3) AC Power, 5 V AC, 0.8 amp

3.3.12.1 Degraded Performance - Degraded performance shall be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions shall be specified.

3.4.1 Operation - The equipment shall be capable of recording either analog and/or digital data signals as selected by the operator. Indicators shall be provided to the operator to monitor the recording of digital and analog data signals. The equipment shall be capable of recording analog data signals upon receipt of a start control signal furnished by an operator actuated foot switch in a manual mode or the presence of audio signals in an automatic mode. All digital signals shall be recorded on the digital tape at the control of step commands generated by external equipment.

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3.4.2 Input Signal Characteristics

3.4.2.1 Digital Data Signals - The following digital data input signals shall be received from external equipment.

*3.4.2.1.1 Data Signals - The digital data signals shall be furnished by external equipment on three parallel lines. The signal level on each of the three lines shall be +5 V (±0.5 V)DC for logic state one and 0.0 (+ 0.6 V, - 0.0 V) DC for logic state zero. The input data lines shall be stable at the desired logic level prior to receipt of the step command and shall remain stable until after the step command has terminated. The equipment shall be capable of receiving up to a maximum rate of 200 characters per second. The input data lines shall be maintained in a logic zero state whenever the digital recorder is being advanced through the longitudinal check character gap and record gap as defined in IBM File No. 729-19, Form A22-6643-3.

*3.4.2.1.2 Step Command - A step command signal shall be furnished by external equipment to initiate the record sequence. One step command shall be generated for each data character described by 3.4.2.1.1 above. One step command shall cause a single data character to be written on the tape in the format described in 3.4.5. The equipment shall be capable of receiving step command signals to maximum rate of 200 steps per second. Each step pulse shall be at a level of +5 V (±0.5 V) DC with a minimum duration of 10 microseconds. The non-step level shall be 0.0 V (+ 0.6 V, - 0.0 V) DC.

*3.4.2.1.3 Clock Command Signal - A clock command signal shall be furnished by external equipment and shall remain set at a level of +5 V (±0.5 V) DC for the duration of the data record. The clock command shall be at 0 V DC (+ 0.6 V, - 0.0 V) DC logic zero state wherever the digital recorder is being advanced through the longitudinal check character gap and the record gap as defined in IBM File No. 729-19, Form A22-6643-3.

*3.4.2.1.4 Time-of-Day Signals - Time-of-day signals shall be furnished by external equipment in binary digital format on 17 parallel lines. The signal format shall be binary hours (five bits), binary

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minutes (six bits) and binary seconds (six bits). A logic one state shall be +4 V (+ 1 V, - 1/2 V) DC and a logic zero state shall be 0 V (+ 0.6 V, - 0.0 V) DC.

3.4.2.1.5 Digitize Indication - A digitize indication signal shall be furnished by external equipment for display by a lighted push-button switch, labeled DIGITIZE, on the control panel. The digitize indication signal shall be 5 V AC and have a minimum duration of 500 milliseconds. This signal shall be furnished by external equipment upon receipt of a digitize command signal from the DIGITIZE pushbutton switch.

*3.4.2.1.6 LRCC Command - A Longitudinal Redundancy Check Character (LRCC) command shall be furnished by external equipment. The LRCC command shall reset the write amplifier flip-flops in the Digital Recorder RO-329/ALH-6 and shall cause a LRCC to be written on the digital tape. The LRCC command shall be stable at a +5 V (±0.5 V) DC logic one level prior to the receipt of the step command associated with the LRCC character, and shall remain stable at the logic one state for at least 10 microseconds after the leading edge of the step command. The LRCC command line shall be at the logic zero state of 0 V (+ 0.6 V, - 0.0 V) DC at all other times.

3.4.2.1.7 Digital Monitor - A digital monitor signal shall be provided on a single wire as an input to the Control, C-7237/ALH-6 to illuminate the Digital Monitor Indicator. This signal shall be generated from the Digital Recorder, RO-329/ALH-6, whenever digital data is being recorded. The signal shall be routed to the Recorder Control Panel via the Analysis Indicator in the Countermeasures Set, AN/ALQ-86, but shall not be used or modified by that equipment in any way.

3.4.2.2 Analog Data Signals -

*3.4.2.2.1 Data Signals - Analog data signals shall be furnished by external equipment on three data lines. The equipment shall be capable of receiving and recording data signals in the bandwidth of 100 to 20,000 cps at signal levels up to 10 V RMS.

3.4.2.2.2 Record Command - A record command signal shall be furnished by external equipment to initiate the analog recorder start circuitry. This signal shall be closure-to-ground furnished by a foot operated switch.

3.4.3 Output Signal Characteristics

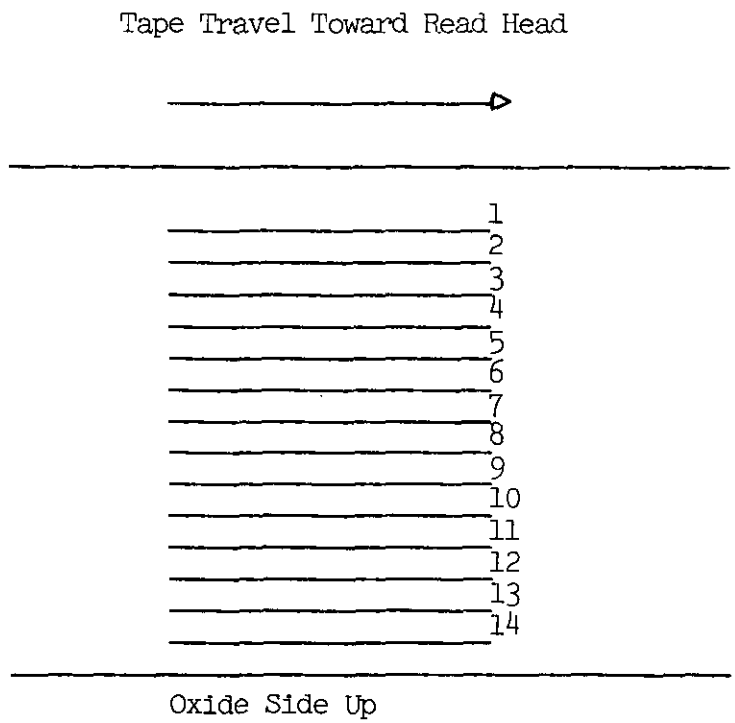
*3.4.3.1 Digital Record - The equipment shall furnish a digital record control signal to external equipment which shall initiate the digital record sequence. The digital record signal shall be negative going pulse from a +5 V (±0.5 V) DC level to a 0 V DC level with a minimum duration of 10 microseconds at the 0 V (+0.6 V, - 0.0 V) DC level.

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*3.4.3.2 Analog Record On - An analog record on signal shall be furnished to external equipment to initiate the digital data sequence to record one record of digital data. The analog record on signal shall be generated any time that the Analog Recorder is started. The analog record signal shall be a negative going pulse from a +5 V (+0.5 V) DC level to a 0 V (+0.6 V, - 0.0 V) DC level with a duration of 10 microseconds minimum at the 0 V (+0.6 V, - 0.0 V) DC level.

3.4.3.3 Digital Monitor - A digital monitor signal shall be generated from the Digital Recorder, RO-329/ALH-6, to indicate that data is being recorded on the digital tape. The digital monitor signal shall be sent from the Digital Recorder to the Recorder Control Panel via the Analysis Indicator in the Countermeasures Set, AN/ALQ-86 (see 3.4.2.1.7).

3.4.4 Analog Tape Format - The analog tape shall be produced in accordance with the following format and channel assignments.

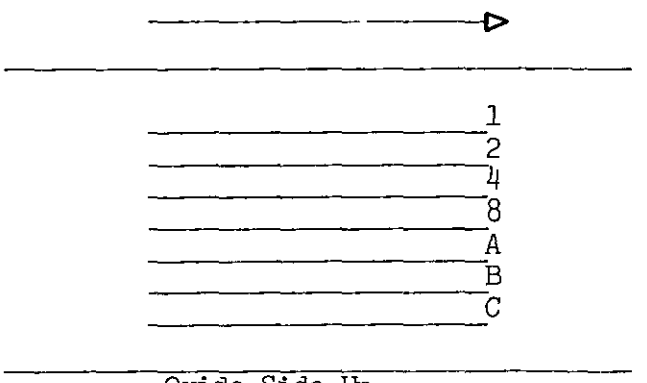


Function	Forward Pass	Reverse Pass
Pilot ICS	Track 1	Track 2
Operator ICS	Track 3	Track 4
1 KC Ref	Track 5	Track 6
Audio Signal	Track 7	Track 8
10 KC Ref	Track 9	Track 10
17 KC Carrier 60 cps am	Track 11	Track 12
Time-of-Day	Track 13	Track 14

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3.4.5 Digital Tape Format - The digital tape shall comply with the IBM File No. 729-19, Form A22-6643, and with the following data track assignments.

Tape Travel Toward Read Head



Track 1 - Data Signal 1
Track 2 - Data Signal 2
Track 4 - Data Signal 3
Track 8 - Data Signal 1
Trace A - Data Signal 2
Track B - Data Signal 3
Track C - Clock Signal

3.4.5.1 Digital Tape Data Fields - The digital tape shall be produced in accordance with the following data field blocking at the control of external equipment. A data field shall consist of 70, seven-bit characters of which the last six characters shall be a field of logical ones to denote end of field. Data fields shall be located sequentially on the tape with no specing between data fields. The bit packing density shall be 200 per-inch.

3.4.5.2 Digital Tape Data Records - The data fields shall be grouped in blocks of 256 data fields placed sequentially on the tape at the control of external equipment with no gaps between the fields. The block shall be followed by an End-of-Record character and a record gap as described by IBM File No. 729-19, Form A22-6643-3. The End-of-Record character and record gap shall be generated at the control of external equipment.

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3.5 Detail Requirements

*3.5.1 Control Recorder C-7237/ALH-6 - The controls shall meet the applicable requirements of MIL-C-6781 for Type 1 control panel.

3.5.1.1 Function - The Control Panel shall provide manual controls and visual indicators pertaining to the operation of the Recorder Reproducer Set. These indicators shall allow the operator to monitor the analog signal recording on an individual track basis and the digital signal recording on a summary basis. Two switches shall be provided on the Control Panel, one to initiate the recording of digital data signals and the other to enable a Voice Operated Relay (VOR) to initiate the recording of analog data signals. In addition, the panel shall provide an indication of recording time elapsed for analog data signals.

*3.5.1.2 Form Factor - The controls, less connectors and front panel, shall be housed in an enclosure no larger than 5.0 inches wide, 3.5 inches high, and 5.0 inches deep. The panel with indicators and switches shall be located on the 5.0 inch and 3.5 inch sides of the enclosure. The maximum external dimensions of the control panel face shall not exceed 5 3/4 inches wide by 3 3/8 inches high.

*3.5.1.3 Weight - The weight of the control and monitor, excluding cables, shall not exceed nine pounds.

*3.5.1.4 Contents - The following assemblies and subassemblies shall be contained in the controls.

Panel with indicators and switches to operate the Recorder-Reproducer Set.

3.5.1.5 Controls - The following controls shall be available.

3.5.1.5.1 Function Controls

3.5.1.5.1.1 Pedal Control Signal - An externally furnished foot switch shall provide the operator a foot-actuated control signal to record analog data signals. This signal shall provide continuous recording of the analog signals during the interval the foot switch is depressed.

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3.5.1.5.1.2 Pilot Record - A panel-mounted control switch labeled pilot record shall be provided to the operator to permit automatic recording of the analog data signals upon the presence of audio signals on the pilot's interior communication signal line. A Voice Operated Relay (VOR) shall be provided within this assembly to start the Analog Recorder when an audio signal is sensed at the input (when the pilot record switch is in the automatic position). The VOR shall remain energized for a period from 1/2 to 2 1/2 seconds after the voice input is removed. An analog recorder start signal shall be generated from this assembly to drive the Analog Recorder at any time the VOR is energized (whenever the pilot record switch is in the automatic position).

3.5.1.5.1.3 Digital Record - A panel mounted illuminated pushbutton labeled DIGITIZE shall be provided to the operator and shall initiate the recording of digital data signals upon actuation by the operator. The pushbutton shall be a momentary action type switch.

3.5.1.6 Indicators - The control panel shall contain the following indicators.

3.5.1.6.1 Monitor Indicators - Indicators lamps shall be provided to monitor data recording stations. Indicators, one for each of seven analog tracks, and one for the Digital Recorder (Digital Monitor), shall be illuminated to indicate data is being recorded. In addition, a panel mounted clock labeled TIME ELAPSED shall be provided to indicate recording time on the analog tape.

*3.5.2 Recorder-Reproducer, Signal Data, RO-330/ALH-6 - The Recorder-Reproducer shall record, store, and read analog signals on magnetic tape with remotely located controls. The Recorder-Reproducer shall meet the following requirements.

3.5.2.1 Function - The equipment shall be capable of recording analog signals simultaneously on seven tracks during the forward pass of the tape using the forward record and reproduce head stack. When the end of the tape is sensed, the logic system automatically reverses tape motion, switches data lines to the reverse record head stack and resumes recording simultaneously on seven other tracks. During the reverse pass of the tape, recording continues until the beginning of the tape is sensed at which time equipment automatically stops the recorder and ignores all additional external record commands. The recorder shall include power supplies, automatic controls, the tape transport, and all logic circuits necessary to allow recording, monitoring, and storing of analog signals on magnetic tape with function controls furnished externally to the equipment.

*3.5.2.2 Form Factor - The magnetic tape Recorder-Reproducer shall be enclosed in a case no larger than 11 inches wide, 5 inches high, and 15 1/2 inches long.

*3.5.2.3 Weight - The weight of the Recorder-Reproducer shall not exceed 30 pounds.

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3.5.2.4 Contents - The Recorder-Reproducer shall contain the following assemblies and subassemblies.

One magnetic tape transport for multi-speed operation

One integral heater with thermostatic switch

One complete set of logic controls as required for the designated mode of operation

One power supply for operation from aircraft AC power

3.5.2.5 Mechanical

3.5.2.5.1 Tape Loading - The Recorder-Reproducer will be used as a tape cartridge; therefore, no tape loading will be necessary in the field.

3.5.2.5.2 Tape Speeds - The tape shall be capable of being driven at speeds of 3.75 or 30 i.p.s.

3.5.2.5.3 Speed Change - Speed change from that stated in 3.5.2.5.2 shall be accomplished as a shop maintenance item by changing belts and pulleys underneath the tape transport.

3.5.2.5.4 Tape - Magnetic tape, one inch wide, with a polyester backing of 0.92 mil plus an oxide coating of 0.18 mil, shall be used to meet the performance specification contained herein. A transparent section of tape shall be provided to serve as the end-of-tape marker at each end of the magnetic tape to initiate automatic sequencing operations.

*3.5.2.5.4.1 Beginning-of-Tape Marker - A transparent section of tape 0.5 inches wide and 5 feet long, beginning two feet from the end of the tape and extending toward the other end shall be provided to serve as the beginning-of-tape marker. This transparent section shall be located in the upper portion of the tape when viewed from the oxide side of the tape.

*3.5.2.5.4.2 End-of-Tape Marker - A transparent section of tape 0.5 inches wide and 5 feet long, beginning three feet from the end of the tape (not at the beginning-of-tape end) and extending toward the other end shall be provided to serve as an end-of-tape marker. This transparent section shall be located in the lower portion of the tape when viewed from the oxide side.

3.5.2.5.5 Reels - Overall dimensions and construction of reels shall be compatible for enclosure within the recorder housing. Reels shall hold not less than 2,250 feet of tape, as specified herein in 3.5.2.5.4.

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3.5.2.5.6 Handling - The Recorder-Reproducer shall be designed to allow manual removal from the aircraft without the need for hand tools. The Recorder-Reproducer shall provide environmental protection for manual handling after removal from the aircraft.

3.5.2.6 Electrical

3.5.2.6.1 Analog Recorder Start - Upon receipt of a 10 V DC (minimum) signal, the recorder shall start and reach the operating speed of 3.75 i.p.s. within 200 milliseconds. The input impedance for this signal shall be 10 K ohms (nominal).

3.5.2.6.2 Ancillary Controls and Indications - The Recorder-Reproducer shall provide electrical circuits, logic and connectors to permit the following additional commands and indicators to interface with external equipment: Fast Forward (30 i.p.s.), Reverse (3.75 i.p.s.) Fast Reverse (30 i.p.s.), Tape Motion, Beginning-of-Tape, and End-of-Tape.

3.5.2.6.3 End-of-Tape Sensing - The equipment shall be provided with end-of-tape sensing. When all the tape has passed from one reel to the other, tape motion shall be automatically reversed.

3.5.2.6.4 Beginning-of-Tape Sensing - The equipment shall be provided with beginning-of-tape sensing. Tape motion shall be automatically stopped upon detection of the transparent section of tape located at the beginning-of-tape.

3.5.2.6.5 Output Data - The Recorder-Reproducer shall provide, analog signals from the read heads for 14 tracks to external equipment to select for monitoring.

3.5.2.7 Performance Requirements - Each Recorder-Reproducer shall operate so as to comply with the performance characteristics described herein.

3.5.2.7.1 Signal to Noise Ratio - The total broadband RMS noise level appearing at the output for any track, recording or playing back at any specified speed into the specified output impedance, shall be less than 25 db below the nominal output signal.

3.5.2.7.2 Crosstalk - The crosstalk between analog tracks shall be at least 40 db less than the nominal recorded signal over the entire signal frequency band when any track is reproduced after the adjacent track or tracks have been recorded simultaneously with the same signals.

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*3.5.2.7.3 Flutter - The steady state flutter appearing on any track shall be less than 2% when recorded at 3 3/4 i.p.s. and measured at a playback speed of 30 i.p.s. under standard conditions and less than 4% at extreme environmental conditions except at vibration resonance frequencies. The flutter shall not exceed 6% at the vibration resonance frequencies.

3.5.2.7.4 Rewind Time - The time required for the complete rewind of tape, from take-up reel full to empty conditions, shall not exceed 15 minutes at 30 i.p.s. for the amount of tape stated elsewhere in this specification.

3.5.2.7.5 Record Time - The total record time shall be not less than 240 min. at a record speed of 3.75 i.p.s., using one full forward pass and one full reverse pass of the tape.

3.5.2.7.6 Speed Regulation - Tape speed shall not exceed a variation of ± 1% under service conditions.

3.5.2.7.7 Control Logic - Circuitry shall be provided to logically control the recorder-reproducer in accordance with control signals provided by the control unit (3.5.1). In addition, beginning-of-tape sensor end-of-tape sensor, tape motion sensor and automatic tape reversal logic shall be provided.

*3.5.3 Converter CV-2278/ALH-6 - The converter shall process signals for recording on the externally connected magnetic tape recorder.

3.5.3.1 Function - The converter shall provide the electronic circuits necessary to process 7 analog data signals for recording by external equipment and shall provide a 750 KC bias signal.

3.5.3.2 Form Factor - The electronics shall be housed in a case no larger than 9.5 inches wide, 2.5 inches high, and 5.0 inches long.

*3.5.3.3 Weight - The weight of the converter shall not exceed five pounds.

3.5.3.4 Contents

One Set of electronic circuits to process data signals.

3.5.3.5 Input Levels - The converter shall have adjustable gain to accomodate signals from 0.5V RMS up to 10 V RMS.

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*3.5.3.6 Input Impedance - The input impedance of the analog data circuits shall be not less than 100 K ohms unbalanced.

3.5.3.7 Frequency Response - Analog recording electronics shall provide the capability to pass signals from 100 cps to 20,000 cps to within ± 3 db.

*3.5.4 Converter - CV-2346/ALH-6 - The converter shall meet the following requirements.

3.5.4.1 Function - The converter shall provide electronic circuits to detect 14 analog signals furnished by external equipment and select one set of 7 signals at a time to monitor and furnish an indicator signal for each data signal to external equipment. Selection of the appropriate seven playback signals shall be accomplished electronically in the playback amplifiers and shall be a function of the direction of "tape travel".

3.5.4.2 Form Factor - The converter shall be housed in an enclosure no larger than 10.0 inches wide, 5.0 inches high, and 8.0 inches long.

3.5.4.3 Weight - The weight of the converter shall not exceed 10 pounds.

3.5.4.4 Contents

1 - Complete set of electronic circuits to detect and select data signals.

3.5.4.5 Controls - The controls for this unit shall be furnished externally to the unit.

*3.5.4.6 Characteristics - The fourteen (14) analog signal lines (3.5.4.1) will be the outputs of the playback heads of the magnetic tape recorder in 3.5.2 operating at 3 3/4 i.p.s. The input impedance of the detectors shall not be less than 10 K ohms. The input switching circuitry shall be controlled by a single +28 (± 0.5) VDC nominal signal that is generated by the recorder. The output turn-on-time shall be within 1.0 ms and shall remain on for sufficient time to ensure that a discernible lamp indication will occur for a 1 ms burst of a 2.5 kc signal.

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*3.5.5 Converter, Signal Data CV-2279/ALH-6 - The Converter shall meet the following requirements.

*3.5.5.1 Function - The converter shall process analog data signals in the 100 to 20,000 cps band furnished by an external equipment. In addition, the converter shall provide three reference signals of 1 kc ± 0.01 percent, 10 kc ± 0.01 percent and 17 kc ± 0.5 percent carrier amplitude modulated (45 to 55 percent) with a 60 cps ± 0.02 percent signal for recording by external equipment. Further, the converter shall provide electronic logic circuitry to reformat 17 bit parallel binary time-of-day signals to 19 bit serial signals consisting of 17 bit binary coded time-of-day signals preceded by a data one (1) signal and a data zero (0) signal. The time-of-day signal shall be shifted serially out of the converter and into the analog recorder. The 1 kc and 2.5 kc reference signals shall be phase locked to the 10 kc reference signal such that the maximum line displacement does not exceed 0.5 microseconds.

3.5.5.2 Form Factor - The converter shall be housed in an enclosure no larger than 6.0 inches wide, 12.0 inches high, and 12.0 inches long.

3.5.5.3 Weight - The weight of the converter shall not exceed 20 pounds.

3.5.5.4 Contents

One complete set of electronic circuitry to process and convert parallel binary time-of-day to serial time-of-day.

One complete set of electronic circuits to process analog data signals.

One complete set of electronic circuitry to provide three reference signals.

One power supply for operation from aircraft AC power.

3.5.5.5 Controls - Controls for this unit will be furnished externally to the unit.

*3.5.6 Recorder - RO-329/ALH-6

3.5.6.1 Function - The Recorder shall include power supplies automatic controls, the tape transport, and all electronic and logic circuits necessary to allow recording and storage of digital signals in the NRZI format described by IBM File No. 729-19, Form A22-6643-3, on magnetic tape with function controls furnished externally to the unit. The Recorder shall be capable of recording digital signals simultaneously

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on seven tracks. Upon receipt of a record step pulse signal from a remote station, the equipment shall process and record data signals on magnetic tape. The Recorder shall output a discrete level to a monitor light on external equipment indicating the recorder is recording. The Recorder shall continue to record data for each step pulse signal until the end-of-tape marker is sensed. Rewind of the tape shall be performed manually at the end of the operation.

3.5.6.2 Form Factor - The magnetic tape Recorder shall be enclosed in a case no larger than 8 inches wide, 4 inches high, and 10 inches long.

*3.5.6.3 Weight - The weight of the Recorder shall not exceed 15 pounds.

3.5.6.4 Contents - The Recorder shall contain the following assemblies and subassemblies.

One magnetic tape transport for incremental operation with provision for plug-in changes of components.

One complete set of logic controls as required for the designated mode of operation

One complete set of electronic circuitry and mechanical components.

One power supply for operation from aircraft AC power

3.5.6.5 Mechanical

3.5.6.5.1 Tape Reel Loading - The equipment shall be provided with the means to enable loading and unloading of tape reels. An interlock shall also be provided to prevent actuation of the equipment in the absence of the reel.

3.5.6.5.2 Manual Threading - The tape shall be manually threaded by wrapping the leader around the take-up spool.

3.5.6.5.3 Record Heads - The construction of the digital heads shall conform to the standards described by IBM File No. 729-19, Form A22-6643-3.

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*3.5.6.5.4 Tape - A magnetic tape, one-half inch wide and 1.5 mil thick, shall be used to meet the performance specification as contained herein. A leader shall be provided at the beginning of the tape to effect manual threading to the transport. A load point and end-of-reel marker shall be affixed at each end of the magnetic tape, in accordance with IBM File No. 729-19, Form A22-6643-3, to initiate the automatic stop sequence.

*3.5.6.5.5 Reels - Reels shall be capable of accepting and storing not less than 600 feet of the tape specified in paragraph 3.5.6.5.4.

*3.5.6.5.6 Reel Hub Adapters - Reel hub adapters shall be constructed with a hub size which is compatible with IBM reel hub assembly (screw type), part number 417944. The reel hub adapter shall be constructed with a groove to provide the write-enable feature.

3.5.6.6 Electrical

3.5.6.6.1 Control - In the record mode of operation, the Recorder shall step the tape and record data upon receipt of a record step signal. The Recorder shall be capable of a maximum rate of 200 steps per second.

3.5.6.6.2 End-of-Tape Sensing - The equipment shall be provided with end-of-tape sensing. Tape motion shall be stopped when all the tape has passed from one reel to the other.

3.5.6.7 Performance Requirements - The Recorder shall operate so as to comply with the performance characteristics described herein.

3.5.6.7.1 Signal Format - The Recorder's signal circuits shall record digital signal inputs on the magnetic tape in accordance with IBM File No. 729-19, Form A22-6643-3.

*3.5.6.7.2 Input Signal Levels - The input signal levels shall be 5.0 (±0.5 V) DC for logic state one and 0.0 (+ 0.6 V, - 0.0 V) for logic state zero.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

*4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. On first production contracts this data shall include a list of major electrical and electronic subassemblies giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

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4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing qualitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

*4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of 10 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only these adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

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Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 10 hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10 hour period specified above may be composed of two 5 hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following: (Sampling tests shall be conducted only when Reliability Assurance Tests of 4.4.3 are deleted by contractual action).

Quantity of Equipments Offered for Acceptance	Quantity to be selected for Sampling Test
First 10	0
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

* NOTE: This test shall not be required if a reliability test is selected that required a burn in test for all equipment.

NOTE: If by contract action the Initial Production Test (4.3) is deleted, a Sample Test shall be conducted on one equipment from the first 10 produced.

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4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

*4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by the Qualification Phase shall be conducted.

*4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of one (1) equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. Test Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the production equipment shall be tested as outlined in MIL-STD-781 (as modified herein), under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase (or All-Equipment Test), Test Level E shall be used.

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4.4.3.2.1 All Equipment Test - Each equipment produced shall be tested for 10 hours. Prior to the 10 hour test on each equipment, an additional burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan IV of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attn: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan IV curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

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4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	27.5 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

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4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of tests procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

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4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

*5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

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6. NOTES

6.1 Intended Use - This equipment shall be used to record on magnetic tape analog and digital data signals which pertain to intercepted electromagnetic emitter signals during airborne EW operations. The equipment will provide magnetic tapes for processing by a ground data readout station.

*6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

*6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

*6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use

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of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

*6.5 Non-Repairable Subassemblies - As a general rule non-repairable, subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

*6.6 Type Designations - The parentheses (*), when used in the type designations, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates shipping records and instruction books, as applicable.

6.7 Revisions - In specification revisions and superseding amendments an asterisk "" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements, of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

*6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

Countermeasures Set - AN/ALQ-86

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-81407A(AS) RECORDER-REPRODUCTER, SIGNAL DATA AN/ALH-6(*)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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